Machines That Accelerate Machine-Making:
A Research Agenda for Human-Demand-Guided Physical Recursive Self-Improvement

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Abstract

Large language models are compressing iteration cycles in software, algorithms, and scientific research, while robot learning, autonomous laboratories, materials discovery, and engineering design carry the same acceleration into the physical world. We argue that this trajectory has a decisive objective: *machines that accelerate machine-making*. We define physical recursive self-improvement (PRSI) as an auditable production relation: after accounting for construction costs, inherited software, inventory, and external capital, an organized physical structure made by the system must still causally improve the production of subsequent improvement operators. This definition distinguishes five causal positions and locates recursion where production capacity itself changes. We then propose *human-demand-guided PRSI*: an outer loop that generates problems, allocates physical search resources, and calibrates forecasts against real outcomes. Formally, we establish the conditional constructibility of finite preregistered certificate chains, geometric contraction of net production loss, a project-selection regret bound under demand-estimation error, contraction of human labor for a fixed product family, and a game-theoretic counterexample in which the system shapes measured demand. These results motivate six organizational capabilities: demand formation, research orchestration, intergenerational memory, physical interfaces, independent metrology, and capital lineage. Our central thesis is that *PRSI lets machines accelerate machine-making, while authorized, plural, outcome-calibrated human demand should guide the direction of that acceleration. With indefinitely extensible recursive chains, declining labor coefficients, and expanding nonlabor capacity, PRSI can open a path to material abundance; when capital gains are inheritable and key complements are controlled, it could become a central arena of competition among individuals, firms, and states.*

Keywords: physical recursive self-improvement; human demand; recursive manufacturing; embodied intelligence; autonomous research and development; industrial competition

1 From LLM Acceleration to Physical Recursion

1.1 The Rate of Improvement Is Becoming Part of Intelligence

Large language models are helping turn capability development from a sequence of occasional breakthroughs into a continuous engineering process. Scaling laws connect performance to data and computation (Kaplan et al., 2020; Hoffmann et al., 2022); pretraining and human feedback extend general task competence (Brown et al., 2020; Ouyang et al., 2022); and tool use together with reasoning–action loops lets models operate directly in software environments (Yao et al., 2023; Schick et al., 2023). Code repositories, test harnesses, and agent runtimes have consequently become environments for AI action. SWE-bench measures changes to real repositories, Voyager accumulates experience as skills, and FunSearch and AI Scientist place program search and research workflows under model control (Jimenez et al., 2024; Wang et al., 2023; Romera-Paredes et al., 2024; Lu et al., 2024).

A longer technical lineage already points toward machines participating in the production of their successors. Artifact optimization automates network architectures, optimization rules, and learning algorithms through neural architecture search, learned optimizers, and AutoML-Zero (Zoph and Le, 2017; Andrychowicz et al., 2016; Real et al., 2020). Open-ended problem generation appears in POET's co-evolution of environments and solutions (Wang et al., 2019). Executable evaluators drive artifact discovery in AlphaTensor, AlphaDev, FunSearch, and AlphaEvolve (Fawzi et al., 2022; Mankowitz et al., 2023; Romera-Paredes et al., 2024; Novikov et al., 2025). Self-editing scaffolds and AI-for-AI training stacks are developed in STOP, the Darwin Gödel Machine, Frontis-MA1, and the Meta-Agent Challenge (Zelikman et al., 2023; Schmidhuber, 2003; Zhang et al., 2025; Yang et al., 2026; Lu et al., 2026). Together, these mechanisms turn the AI-GA vision of processes that generate successor intelligence (Clune, 2019) into software research loops that can be programmed, executed, and compared (Anthropic Institute, 2026). Their unresolved physical boundary is inheritance: software artifacts can alter the next search, while machine capital must also be built, qualified, and causally credited.

Software loops depend on chips, sensors, actuators, machine tools, metrology, energy, and supply chains. When AI rewrites code, it improves the use of existing machines. When AI improves the means by which successor machines are manufactured, calibrated, maintained, and validated, it improves production capacity itself. The latter transition defines physical recursive self-improvement.

1.2 Five Causal Positions Through Which AI Enters the Physical World

Hardware interfaces, robot learning, real-world experience, autonomous R&D, and physical recursion occupy different causal positions. Table 1 classifies them by asking what AI ultimately changes. The labels P1–P5 denote five positions rather than a strictly serial maturity ladder: the policy-experience loop at P3 can proceed in parallel with product R&D at P4, while P5 can combine P1–P4 or close a production operator directly through mechanical self-replication and reconfigurable manufacturing.

Table 1: Five causal positions through which AI reaches the physical world.

Class	Object of change	Representative research lineages	Causal boundary
P1	Hardware interfaces	Device protocols, code policies, model–hardware interfaces	AI gains device read/write access
P2	Policies for a fixed morphology	Control, demonstration, reinforcement learning, Sim2Real, diffusion policies, VLA	Policy changes; body and production line are inherited
P3	Real-world experience loop	Physical interaction, automatic reset, online adaptation, self-generated data	Physical outcomes continually update policy
P4	Product and process R&D	Robot scientists, self-driving laboratories, materials and device design	R&D accelerates; production capital is inherited
P5	Operators that produce successor machines	Self-replication, reconfigurable manufacturing, closure with autonomous R&D	Output improves the capacity to produce successor machines

The P1 interface lineage includes ROS 2, SiLA 2, Code as Policies, ProgPrompt, and emerging model–hardware interfaces (Macenski et al., 2022; Juchli, 2022; Liang et al., 2023; Singh et al., 2023; Anthropic, 2026; dimensionalOS Contributors, 2026). P2 spans control theory, learning from demonstration, reinforcement learning, domain randomization, diffusion policies, and robotic foundation models; VLA is one branch that connects multitask semantics to action generation (Mayne et al., 2000; Argall et al., 2009; Kober et al., 2013; Tobin et al., 2017; OpenAI et al., 2019; Chi et al., 2023; Driess et al., 2023; Brohan et al., 2023; Kim et al., 2024;

Black et al., 2024). P3 brings physical interaction, reset, and verification into continual learning (Kalashnikov et al., 2018; Sharma et al., 2022; Bousmalis et al., 2023; Hirose et al., 2024; Xiao et al., 2026). P4 advances the same closed-loop logic into biology, chemistry, materials, chip design, and nanophotonics (King et al., 2004; Häse et al., 2019; Burger et al., 2020; Szymanski et al., 2023; Boiko et al., 2023; Merchant et al., 2023; Mirhoseini et al., 2021; Molesky et al., 2018).

P5 shifts the object of improvement to production capacity itself. Self-replicating machines and reconfigurable manufacturing establish principles for replication, assembly, and the reorganization of capacity (von Neumann, 1966; von Tiesenhausen and Darbro, 1980; Freitas et al., 1982; Moses and Chirikjian, 2020; Koren et al., 2018). Strict PRSI additionally requires those outputs to enter the next cycle of machine development and to advance the production frontier under full-cost accounting and causal controls.

1.3 From Cumulative Culture to Human-Demand Guidance

The distinctive power of human toolmaking lies in cumulative inheritance. Technologies can be preserved, evaluated, recombined, and improved until they embody capabilities that no individual could invent alone (Tomasello, 1999; Zwirner and Thornton, 2015; Brinkmann et al., 2023). Variation, retention and transmission, and social selection jointly shape this process; memory, learning, and selection biases determine which tools reach the next generation (Thompson and Griffiths, 2021). The growth of tools therefore poses two questions: how can we make better things, and which things deserve to be carried forward?

The second question brings human demand into PRSI. Demand observations translate social selection in cumulative culture into explicit and revisable engineering signals. Technical bottlenecks act as focusing devices, market and organizational demand allocate innovative effort, and lead users often expose problems and create prototypes before mass markets recognize them (Rosenberg, 1969; Mowery and Rosenberg, 1979; von Hippel, 1986). Power, imitation, and institutional lock-in also influence selection, so demand signals require authorized observation across groups and calibration against real outcomes. Such signals encompass friction in production and daily life, public values, and long-term capabilities. They supply a problem distribution for open-ended physical search, while adoption, rejection, failure, and welfare outcomes determine which tools gain opportunities to replicate.

Self-reproducing automata explain how a machine can copy itself. Reconfigurable manufacturing explains how capacity can change. Autonomous laboratories explain how experiments can proceed continuously. PRSI adds a production-relation criterion:

> **PRSI is machines accelerating machine-making.** A system achieves recursive improvement in the physical sense only when the physical organization it produces—after accounting for construction costs, inherited software, inventory, and external capital—still causally improves the production of subsequent improvement operators.

1.4 Five Stages of Human-Demand Guidance

Table 2 describes how deeply demand enters an improvement loop. Higher stages expand problem discovery and forecasting, while real outcomes continually recalibrate the model.

Table 2: Five stages through which human demand enters an improvement loop.

Stage	Demand representation	Typical mechanism	Principal capability boundary
D1	Fixed task or engineering metric	Preset reward, specification, and test	Demand remains static
D2	Feedback on outputs or trajectories	Preference learning and RLHF	Feedback is local and reactive
D3	Stable principles or constitution	Constitutional AI	Principles remain stable while contexts change
D4	Online belief over demand	CIRL, active querying, continual personalization	Coverage is limited by interaction samples
D5	Large-scale models of heterogeneous and future demand	Generative agents, specialized user models, behavioral and population simulation	Forecasts require calibration in a real-outcome loop

Learning from feedback and constitutional principles support D2–D3 (Christiano et al., 2017; Ouyang et al., 2022; Bai et al., 2022; Kundu et al., 2023). Cooperative reward learning, continual personalization, and user simulation support D4–D5 (Hadfield-Menell et al., 2016; Liang et al., 2026; Park et al., 2023; Yoon et al., 2024; Naous et al., 2025; Wu et al., 2026).

The resulting claim chain is compact. Physical production edges define PRSI. When demand-estimation error is controlled and selection quality is tested against real outcomes, demand models can improve the allocation of scarce R&D resources. A coupled-loop architecture then connects demand, research, manufacturing, measurement, and reinvestment, with implications for material abundance, strategic competition, and the co-evolution of demand and productive capacity.

2 Recursive Improvement Becomes Physical When Machines Change the Means of Making Machines

2.1 Dual Inheritance: Informational State and the Built World

The successor state of a software system is determined chiefly by code, parameters, data, and the runtime environment. A physical system also inherits a built world: machine tools, fixtures, sensors, qualified processes, inventories, calibration, wear states, energy interfaces, and custody relations. Let the production state of generation k be

$$P_k = (S_k, W_k, R_k, Z_k), \tag{1}$$

where S_k is a transferable information strategy, W_k is the built world, R_k comprises budget, operating conditions, and governance constraints, and Z_k records evidence and lineage. We call this decomposition *dual inheritance*. Information can improve how machines are used; physical organization can improve how machines are produced. Both enter the successor state, but their effects must be attributed separately.

Dual inheritance requires separate accounting for models, inference budgets, and human-authored scripts on one side, and procured equipment, prefabricated inventory, and maintenance inputs on the other. When firmware and physical components form a composite, the estimand is the treatment effect of that composite as a whole.

2.2 The Physical Improvement-Production Edge

For candidate edge $k \geq 1$, let t_k be the actual input-operator instance used in the registered production event, δ_k the physical production transition within an authorized safety envelope, and o_k the qualified output operator. Write $\mathsf{ProducedBy}(o_k; P_{k-1}, t_k, \delta_k)$ when registered execution and custody records establish that t_k was actually invoked through δ_k to produce and qualify

o_k from ancestor state P_{k-1}. The resource vector $\mathbf{c}(P;\mu)$ records the time, materials, energy, money, labor, and failure loss required for state P to produce a qualified successor operator on task distribution μ. Quality, safety, rights, and environmental limits define the feasible set $\mathcal{F}$. Within that set, a preregistered rule compares net production loss, $\ell(P;\mu) = \omega^\top \mathbf{c}(P;\mu)$, where $\omega \succeq 0$. The design, fabrication, calibration, validation, and reset costs of a candidate tool all enter $\mathbf{c}$.

The causal comparison generates two potential successor states from the same ancestor P_{k-1}. The treatment fork, $P_k^{(1)} = \mathsf{Fork}(P_{k-1}; o_k)$, uses the candidate physical organization. The materially equivalent control fork, $P_k^{(0)} = \mathsf{Fork}(P_{k-1}; \varnothing)$, does not use its organizational function. The forks bind software and runtime, initial inventory, external capital, maintenance allowance, task-revelation schedule, and evaluator; the only preregistered difference is the organizational function of o_k. Define

$$\tau_k(\mu) = \mathbb{E}\Big[\widehat{\ell}_T(P_k^{(1)};\mu) - \widehat{\ell}_T(P_k^{(0)};\mu)\Big] \tag{2}$$

as the causal effect on net production loss under a common ancestor. After the treatment fork is accepted, write the successor state as $P_k := P_k^{(1)}$.

Definition 1 (Physical improvement-production edge)**.** *Let a candidate edge be*

$$e_k = (P_{k-1}, t_k, \delta_k, o_k, P_k). \tag{3}$$

*It is a physical improvement-production edge (*PIPE*) if and only if all of the following conditions hold:*

1. *δ_k is completed within the registered authority, budget, and safety envelope, and $\mathsf{ProducedBy}(o_k; P_{k-1}, t_k, \delta_k)$ holds;*
2. *the materially equivalent forks bind every non-treatment variable, identifying the organizational function of o_k rather than a change in software, inventory, or capital;*
3. *for a preregistered minimum material effect $\varepsilon_k > 0$, $\tau_k(\mu_k) \leq -\varepsilon_k$, and the treatment fork remains in $\mathcal{F}$ on every noncompensable outcome;*
4. *software, inventory, external capital, identity, and custody flows are recorded completely; and*
5. *on hidden inputs, o_k repeatedly changes the production process and therefore supplies a generalizable production function.*

Common-ancestor accounting returns every fork to the same starting point. For a registered task distribution μ and a fixed, normalized horizon of $T \geq 1$ uses, let $L_{\text{inc-build}}(P \mid P_{k-1})$ denote the incremental cost of designing, building, and qualifying the candidate organization beyond the common ancestor, rather than the cost of the entire inherited state. Let $L_{\text{capital},i}$ charge inherited-capital depreciation and opportunity cost under the same preregistered valuation schedule in both forks. One operational estimator of net loss is

$$\begin{aligned}\widehat{\ell}_T(P;\mu) = \frac{1}{T}\Bigg[L_{\text{inc-build}}(P \mid P_{k-1}) + \sum_{i=1}^{T}\Big(&L_{\text{plan},i} + L_{\text{procure},i} + L_{\text{fabricate},i} \\ &+ L_{\text{calibrate},i} + L_{\text{verify},i} + L_{\text{reset},i} \\ &+ L_{\text{quality},i} + L_{\text{capital},i} + \lambda_{\text{f}}\mathbf{1}\{\text{failure}_i\}\Big)\Bigg].\end{aligned} \tag{4}$$

Task i is drawn from μ, $L_{\text{quality},i}$ measures residual loss inside the feasible set, λ_{f} penalizes failure, and $\ell(P;\mu) = \mathbb{E}[\widehat{\ell}_T(P;\mu)]$. Randomized or paired forks estimate Eq. (2); its one-sided confidence bound must lie below $-\varepsilon_k$. Both groups share the task distribution, horizon, inventory, runtime, and revelation schedule. A fixture advances the net production frontier only if it repays its build cost within the registered horizon. Work completed in advance and software exclusive to the treatment group are charged to the corresponding fork. Treatment, control, intervention timing, and potential confounders follow standard principles of causal inference (Pearl, 2009; Hernán and Robins, 2020).

This boundary separates P3 from P5. Physical interaction, automatic reset, online adaptation, and self-generated data can continually improve a policy (Kalashnikov et al., 2018; Sharma et al., 2022; Bousmalis et al., 2023; Hirose et al., 2024; Xiao et al., 2026). If the system then designs and fabricates a reset or metrology device, and that device improves subsequent policy development under a net-cost control, the device forms a candidate PIPE. Recursion resides in the production operator that has changed.

2.3 Constructing a Finite Preregistered Certificate Chain

Once finite production edges become the unit of claim, we can ask a direct question: given discoverable positive edges and a decidable certificate at every generation, can a finite certificate lineage be constructed mechanically? After fixing the task set, horizon T, confidence level, preregistered thresholds, multiplicity correction, and policy for adaptive candidate search, let $\mathsf{PIPE}_T(e)$ mean that the finite evidence package for edge e passes the authorization, production-identity, feasibility, cost, and selection-adjusted statistical tests associated with Eq. (2). In particular, the package must certify $\mathsf{ProducedBy}$. A verifier V_T inspects only this finite certificate. Define

$$\mathsf{Chain}_{n,T}(e_1,\dots,e_n) \iff \left(\bigwedge_{i=1}^{n} \mathsf{PIPE}_T(e_i)\right) \wedge \left(\bigwedge_{i=1}^{n-1} o_i = t_{i+1}\right). \tag{5}$$

Theorem 1 (Conditional constructibility of a finite preregistered certificate chain)**.** *Let $\mathcal{S}$ be a nonempty set of safely reachable states, with $P_0 \in \mathcal{S}$. The initial candidate set $\Delta_0(P_0)$ is finite and enumerable and contains at least one witness satisfying PIPE_T. For every verified prefix $C_k = (e_1,\dots,e_k)$ whose terminal state is $P_k \in \mathcal{S}$ and terminal output operator is o_k, suppose the extension set $\Delta(P_k, o_k)$ is finite and enumerable; every candidate e_{k+1} in it has input identity fixed to $t_{k+1} = o_k$ and must certify $\mathsf{ProducedBy}(o_{k+1}; P_k, t_{k+1}, \delta_{k+1})$; and the set contains at least one witness satisfying PIPE_T. Suppose further that the registered protocol controls multiplicity and adaptive search over every inspected candidate, that V_T is sound, complete, and terminating for this finite predicate, and that the terminal state of every accepted edge remains in $\mathcal{S}$. Then a computable process constructs a preregistered certificate chain satisfying $\mathsf{Chain}_{n,T}$ for every finite $n \geq 1$.*

Proof. Enumerate $\Delta_0(P_0)$ and run V_T. Finite enumeration, termination of verification, and the existence of a positive witness guarantee that the search accepts a candidate in finite time; soundness guarantees that the resulting e_1 satisfies PIPE_T. Now assume a prefix C_k has been constructed with terminal state $P_k \in \mathcal{S}$ and terminal output o_k. Enumerating $\Delta(P_k, o_k)$ and running V_T similarly yields an edge e_{k+1}. The candidate set enforces $t_{k+1} = o_k$, while the accepted PIPE_T certificate establishes $\mathsf{ProducedBy}(o_{k+1}; P_k, t_{k+1}, \delta_{k+1})$ and closure keeps the new terminal state inside $\mathcal{S}$. Thus C_k extends to the exactly linked C_{k+1}, and induction proves the claim. □

Theorem 1 reduces a preregistered certificate chain to finite candidate search, a positive witness at each generation, a decidable statistical protocol, actual operator use, and closure of the safe set. The predicate $\mathsf{Chain}_{n,T}$ is a finite protocol-acceptance record. It neither entails the real-chain predicate Chain_n below nor guarantees another edge. Under stated task-sampling assumptions, selection-adjusted confidence control, and independent replication, it provides evidence for—but does not deductively establish—a real-chain claim. The central engineering problem of PRSI remains the continual creation of real positive edges.

Proposition 1 (Conditional geometric contraction of net production loss)**.** *Fix a nonempty anchor task distribution* μ^**, a common normalized horizon, and a nonnegative loss* ℓ*. If a real chain satisfying* Chain_n *admits a constant* $0 < \rho < 1$ *such that*

$$\ell(P_{k+1}; \mu^*) \leq \rho \ell(P_k; \mu^*), \qquad k = 0, \ldots, n-1, \tag{6}$$

then $\ell(P_n; \mu^*) \leq \rho^n \ell(P_0; \mu^*)$.

Proof. Repeated substitution yields $\ell(P_n; \mu^*) \leq \rho \ell(P_{n-1}; \mu^*) \leq \cdots \leq \rho^n \ell(P_0; \mu^*)$. □

Proposition 1 gives a strong condition for accelerated machine-making: under a fixed anchor and accounting rule, net production loss declines by a constant proportion at every generation. Acceleration in wall-clock time additionally requires the time component to contract. Additive gains or convergence under resource limits can still create PRSI edges, but they do not imply geometric contraction. A growth model should represent electricity, data, bandwidth, and capital constraints explicitly (Jafari et al., 2025).

2.4 Exactly Linked Chains and Finite Certificates

Let o_i denote the identity of the output operator produced by edge e_i, and let t_i denote the actual input-operator instance used by that edge. Define

$$\mathsf{Chain}_n(e_1, \ldots, e_n) \Longleftrightarrow \left(\bigwedge_{i=1}^{n} \mathsf{PIPE}(e_i) \right) \wedge \left(\bigwedge_{i=1}^{n-1} o_i = t_{i+1} \right). \tag{7}$$

Together with the $\mathsf{ProducedBy}$ conjunct inside each PIPE, the identity condition requires generation two actually to use the same fixture instance A made by generation one in its registered transition. A procured fixture B with the same function can serve in a materially equivalent control, but it belongs to a different lineage.

Proposition 2 (Prefix closure and non-extrapolation of finite chains)**.** *If* Chain_n *holds, then for every* $1 \leq m \leq n$*, its prefix of length* m *satisfies* Chain_m*. Yet no finite* Chain_n *entails the existence of an edge* e_{n+1} *that extends it.*

Proof. Prefix closure follows by deleting terminal conjuncts from the finite conjunction in Eq. (7). For non-extrapolation, construct a model whose edge domain contains exactly $e_1, \ldots, e_n$, with every existing PIPE predicate and adjacent identity equation true. Then Chain_n is true, while the domain contains no witness for e_{n+1}, so the extension claim is false. □

The scientific program of PRSI is therefore to produce auditable edges one by one, while studying the physical, economic, and organizational mechanisms that raise the probability of the next edge. The next section introduces human demand and asks how a system should choose which candidate edges deserve to be attempted.

3 How Human-Demand Models Can Guide PRSI

3.1 Open-Ended Improvement Begins with Problem Selection

Materials, robot time, expert attention, and validation facilities are scarce. Problem selection determines where a powerful solver spends these resources. Open-ended learning shows that a problem distribution can itself generate selection pressure and stepping stones; POET drives capability growth by co-evolving environments and solutions (Wang et al., 2019). For PRSI, the friction that people encounter in production and daily life is a real-world problem generator.

Demand is evidence about problems, stakeholders, constraints, and desired outcomes. It includes worker workarounds, accessibility barriers, equipment hazards, calibration costs, public maintenance bottlenecks, and observed adoption or rejection. Value-sensitive design and participatory AI bring affected parties into the definition of problems, rights, and criteria for success (Friedman and Hendry, 2019; Delgado et al., 2023). Market, organizational, and public values enter the outcome vector together, while safety, rights, and consent define its feasible set.

Agent evaluation already organizes users, tools, policy constraints, and stateful outcomes as an interactive process (Yao et al., 2024). Human-demand-guided PRSI goes further: it discovers problems from workflows, failures, workarounds, and unmet aspirations; predicts which problems merit physical R&D; and tests both human outcomes and effects on successor production capacity. Demand discovery thus becomes an upstream component of production.

3.2 A Two-Stage Loop: Demand Formation and Demand Resolution

The minimal system has two consecutive stages with separate authority over factual judgments.

Stage one: demand formation. The system forms candidate problems from authorized interviews, workflows, failure records, behavior, and domain knowledge. Simulated users or populations of personas provide formative feedback on proposals, exposing ambiguity, tail cases, and counterfactual responses. A demand analyst then revises and freezes the brief before execution: target population, observation window, candidate actions, predicted outcome distribution, constraints, model identity, and evaluation plan.

Stage two: demand resolution. The frozen brief enters a physical R&D system. The solver proposes designs, processes, fixtures, metrology, or machine components; calls simulation and hardware; and submits product utility and production-capacity effects to independent evaluators. The resulting evidence returns to the demand model and updates its belief about which problems merit attention. If an output also serves as a successor production operator, it enters a PIPE test.

3.3 User Simulation Expands the Search over Demand

Large language models allow demand formation to operate at greater scale. Generative agents maintain memory, relationships, and daily behavior (Park et al., 2023). User simulation now supports the evaluation of conversational recommenders, tool-using agents, and multi-turn assistants (Yoon et al., 2024; Yao et al., 2024; Dou et al., 2025). Specialized user language models, latent-state alignment, grounding in real behavior, and heterogeneous-behavior benchmarks turn the "user" from a short role prompt into a trainable, comparable, and calibratable system component (Naous et al., 2025; Wu et al., 2026; Zhu et al., 2026; Hu et al., 2026). Together, these research lines provide a technical basis for demand exploration at scale.

The fidelity of a user model depends on the task, grounding method, population coverage, and persistence of state. SimulatorArena compares profile-conditioned simulators using human judgments, while Sim2Real studies directly measure the gap between simulated users and real people (Dou et al., 2025; Zhou et al., 2026). A simulator should therefore generate candidate problems, apply diverse stress, and commit to falsifiable forecasts; real outcomes update and select the model.

Let h_t be the authorized context at decision time t, A_t the candidate action set, and D_ϕ the demand model. Before execution, the model submits

$$\widehat{p}_{\phi,t}(y \mid a, h_t), \qquad a \in A_t. \tag{8}$$

After action a_t begins, an independent adapter records the real outcome y_t over the registered window. Categorical outcomes use a Brier score; continuous outcomes use a preregistered loss.

Payment, deployment, repeated use, failure, harm, rejection, and censoring are recorded as distinct events, and forecast residuals enter the next model update.

3.4 How Demand Guidance Improves Project Selection: An Error Bound

Demand models can support PRSI by improving the allocation of a finite R&D budget. First define a nonempty finite feasible set through hard limits on rights, safety, consent, and resources:

$$A_{\text{safe}} = \{a \in A : \mathsf{Rights}(a) \wedge \mathsf{Safety}(a) \wedge \mathsf{Consent}(a) \wedge \mathsf{Budget}(a)\}. \tag{9}$$

For every $a \in A_{\text{safe}}$, define its true coupled value as

$$J(a) = \alpha U(a) + \beta B(a) - C(a), \tag{10}$$

where $U(a)$ is the value of authorized human outcomes, $B(a)$ is the gain in physical production capacity, $C(a)$ is common-ancestor cost, and $\alpha, \beta > 0$ are preregistered weights. The system estimates $\widehat{U}$, $\widehat{B}$, and $\widehat{C}$, then selects the project that maximizes $\widehat{J}$.

Theorem 2 (Selection regret under uniform estimation error)**.** *Suppose that for every $a \in A_{\text{safe}}$,*

$$|\widehat{U}(a) - U(a)| \le \varepsilon_U, \quad |\widehat{B}(a) - B(a)| \le \varepsilon_B, \quad |\widehat{C}(a) - C(a)| \le \varepsilon_C. \tag{11}$$

Let $\Delta = \alpha\varepsilon_U + \beta\varepsilon_B + \varepsilon_C$, and define $\widehat{a} \in \arg\max_{a \in A_{\text{safe}}} \widehat{J}(a)$ and $a^* \in \arg\max_{a \in A_{\text{safe}}} J(a)$. Then

$$J(\widehat{a}) \ge J(a^*) - 2\Delta. \tag{12}$$

If $|A_{\text{safe}}| \ge 2$, a^* is unique, and $J(a^*) - \max_{a \ne a^*} J(a) > 2\Delta$, then the estimated ranking selects a^*.

Proof. The triangle inequality gives $|\widehat{J}(a) - J(a)| \le \Delta$ for every a. Hence

$$J(\widehat{a}) \ge \widehat{J}(\widehat{a}) - \Delta \ge \widehat{J}(a^*) - \Delta \ge J(a^*) - 2\Delta. \tag{13}$$

If $\widehat{a} \ne a^*$, estimation error can reverse only a pair whose true gap is at most 2Δ. A larger true gap yields a contradiction, so $\widehat{a} = a^*$. □

Theorem 2 directly concerns coupled project selection: reducing ε_U tightens worst-case regret on a fixed candidate set. Transmission to physical recursion requires a testable bridge—lower selection regret must raise the discovery rate of positive PIPEs, physical benefit, or recursive progress per unit budget. Under the same physical budget, an experiment can compare expert rules, a general-purpose LLM, a calibrated user model, and a hybrid committee. The outcomes should include held-out J, selection regret, PIPE hit rate, and net production benefit as distinct measures.

3.5 Active Updating Matters More Than Static Profiles

Users learn new products, organizations rearrange work, and the system creates new possibilities. Cooperative inverse reinforcement learning models human–machine interaction as a cooperative game under uncertainty about human reward, producing active learning, active teaching, and communication behavior (Hadfield-Menell et al., 2016). Continual personalization adapts to preference drift through clarification before action, memory retrieval, and feedback after action (Liang et al., 2026). D4 therefore decides when to ask, whom to ask, and how to communicate uncertainty.

D5 forecasts future demand, but its forecasts change products, markets, and the distribution of preferences. The demand loop is both the steering mechanism for PRSI and a channel through which production capacity and human demand co-evolve.

4 Machines That Accelerate Machine-Making: Conditions for System Construction

Physical edges and demand-based selection form human-demand-guided PRSI only when they enter the same inheritable production chain. A long-lived R&D organization must bind predictions, execution, measurement, cost, and reinvestment to common identities and timelines, while preserving separate procedures for establishing each kind of fact.

4.1 Two Loops, One Production Chain

Figure 1 shows the coupled architecture. The demand loop maintains a distribution over what is worth making. The physical loop maintains the capacity to make machines better. The former sends frozen briefs; the latter returns product outcomes, costs, failures, and effects on successor production capacity.

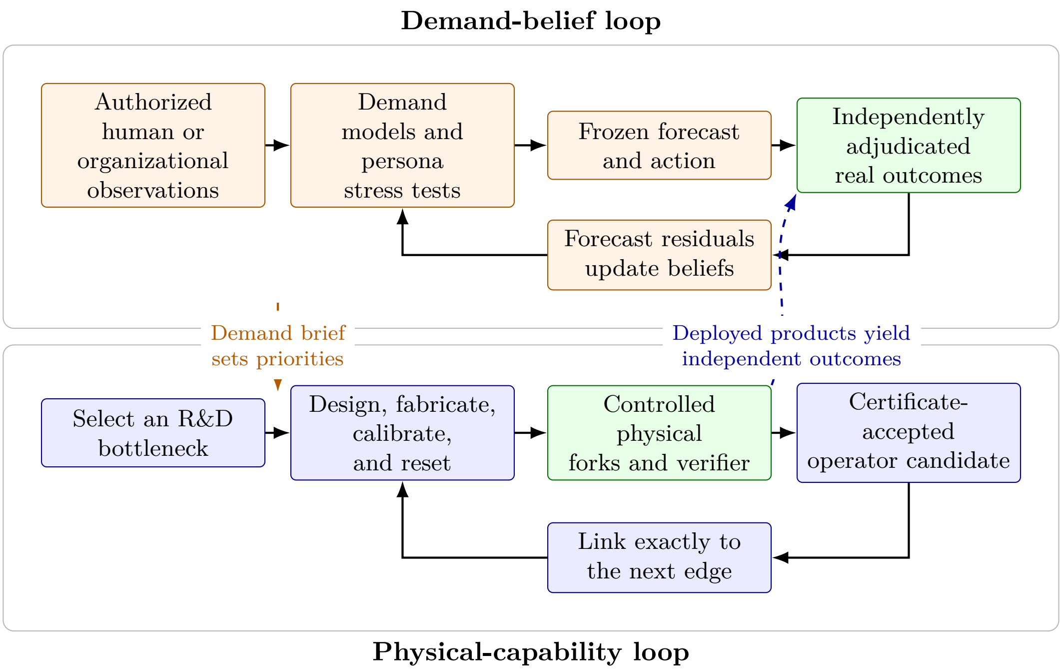

Figure 1: The coupled PRSI proposition at the certificate level. Simulated users and demand models shape the search distribution over physical problems. Authorized real-outcome procedures adjudicate product and deployment effects, while controlled physical forks adjudicate registered evidence about recursive capability. Recursive attribution requires a certificate-accepted operator candidate to be used in the next edge. The diagram depicts protocol-level acceptance, not proof of Chain_n. Orange denotes guidance, blue denotes physical capability, green denotes authority held independently by an evaluator, and dashed arrows denote cross-loop influence.

A demand episode can be represented as

$$q_t = (h_t, A_t, \widehat{p}_t, a_t, u_t, y_t, m_t, \tau_f, \tau_x, \tau_y), \tag{14}$$

where $h_t, A_t, \widehat{p}_t, a_t, u_t, y_t, m_t$ denote, respectively, the authorized context, frozen action set, ex ante outcome forecast, executed action, outcome evaluator, observed outcome, and provenance–consent–custody metadata. Freezing, execution, and observation obey $\tau_f < \tau_x \leq \tau_y$. A physical edge e_k separately binds causal forks, operators, resources, evaluators, and custody. Let $C_n = (e_1, \ldots, e_n)$. Selection, product outcome, and recursive position can then be connected by three relations: $\mathsf{Directed}(q_t, e_k)$, $\mathsf{OutcomeOf}(q_t, \mathsf{product}(e_k))$, and $\mathsf{Last}(e_k, C_n)$.

This separation yields a registered evidence gate:

$$\begin{aligned}\mathsf{Accept}_T(q_t, e_k, C_n) \iff & \mathsf{AuthorizedOutcome}(q_t) \wedge \mathsf{Directed}(q_t, e_k) \\ & \wedge \mathsf{OutcomeOf}(q_t, \mathsf{product}(e_k)) \wedge \mathsf{Last}(e_k, C_n) \\ & \wedge \mathsf{Chain}_{n,T}(C_n) \wedge \mathsf{Safe}(q_t, e_k).\end{aligned} \tag{15}$$

The conjunction in Eq. (15) preserves two classes of registered fact. Authorized observation establishes human outcomes; a finite certificate lineage records protocol-level evidence about physical capability. The certificate predicate neither entails Chain_n nor guarantees a successor edge. A real PRSI claim separately requires the Chain_n predicate in Eq. (7), supported under explicit sampling assumptions, selection-adjusted confidence control, and external replication.

4.2 Six System Capabilities

Equation (15) can be operationalized as six composable capabilities:

1. **Demand formation and precommitment:** form candidate problems from real friction, then freeze the population, actions, budget, forecasts, and success conditions before fabrication.

2. **Research orchestration and design compilation:** continuously compile demand into retrieval, modeling, simulation, design, procurement, fabrication, and validation workflows.

3. **Intergenerational R&D memory:** version designs, experiments, failures, calibration, software, and evidence so that successor research inherits organizational knowledge (Teece et al., 1997; March, 1991).

4. **Physical execution interfaces:** connect instruments, robots, and machine tools through addressable and recoverable primitives, binding telemetry to action identity (Macenski et al., 2022; Juchli, 2022; Liang et al., 2023; Anthropic, 2026; dimensionalOS Contributors, 2026).

5. **Independent metrology and outcome adjudication:** measure human outcomes, product quality, and production capacity separately through predetermined procedures.

6. **Resource lineage and capital reinvestment:** track identity, inventory, software, external capital, and common-ancestor cost, then turn verified gains into the next generation of machine capital.

4.3 A General Implementation Layer

These capabilities can reside in workflow engines, versioned knowledge bases, CAD/CAE and experimental toolchains, device-adapter layers, independent measurement services, lineage databases, and financial systems. Implementation boundaries follow the movement of information through demand, design, manufacturing, validation, and reinvestment, while allowing models, software, and equipment to be replaced over time.

4.4 From a Finite Seed to a PRSI Organization

The logical carrier of these six capabilities is an organization that persistently holds capital, knowledge, responsibility, and outcome records. A firm is one clear implementation, but a cooperative, public laboratory, or interinstitutional alliance could close the same loop. In a PRSI organization, demand sensing, R&D exploration, physical realization, and capital reinvestment form a continuous machine process (Teece et al., 1997; March, 1991).

A minimal seed can begin with a narrow, frequent production bottleneck. The system identifies the cost of machine commissioning, designs and fabricates a calibration device, measures its net benefit across multiple subsequent machine-development episodes, and then uses that device to produce the next generation of metrology or assembly equipment. One PIPE, a second edge linked by exact operator identity, and the progressive internalization of materials, energy, maintenance, and supply form a path from an engineering prototype to recursive production capacity.

5 Abundance, Competition, and the Game over Demand

PRSI turns an industrial system into an entity that can learn, manufacture, and inherit new productive capabilities. It changes material output, ownership of the production frontier, and the joint evolution of machines and human demand.

5.1 Material Abundance and Liberation from the Necessity of Labor

The long-run promise of PRSI is material abundance. Automation displaces some tasks while creating new tasks and maintenance needs (Acemoglu and Restrepo, 2019). A recursive system continues by improving automated machines, the machines that make them, and the equipment that validates them. When human labor coefficients in design, manufacturing, calibration, maintenance, and quality control decline across generations while nonlabor capacity expands, the next production cycle also becomes less dependent on labor. Keynes described emancipation from economic necessity as the long-run objective of technological progress (Keynes, 1931). PRSI supplies an engineering mechanism for that vision.

The mechanism admits a direct limit statement. For a fixed basket of qualified products b, let $h_k(b) > 0$ be the expected human labor hours required to produce one unit of b in state P_k, including design, operation, maintenance, calibration, and validation.

Proposition 3 (Contraction of labor demand for a fixed product family)**.** *Suppose an indefinitely extensible PRSI chain admits $0 < \rho_h < 1$ such that*

$$h_{k+1}(b) \le \rho_h h_k(b), \qquad k \ge 0, \tag{16}$$

and nonlabor inputs suffice to sustain a fixed demand D. Then required human labor, $H_k(D, b) = Dh_k(b)$, converges to zero. Further, let $Q_k^{\mathrm{H}}(H, b) = H/h_k(b)$ be labor-constrained output, and let $\overline{Q}_k^{\mathrm{NL}}(b)$ be the maximum output permitted jointly by materials, energy, and machine capacity. Assume that both quantities describe the same production technology, that the constraints are separable, and that no other bottleneck exists, so joint feasible output is

$$Q_k^{\mathrm{actual}}(H, b) = \min\{Q_k^{\mathrm{H}}(H, b), \overline{Q}_k^{\mathrm{NL}}(b)\}. \tag{17}$$

If, for fixed labor $H > 0$,

$$\overline{Q}_k^{\mathrm{NL}}(b) \ge \frac{H}{h_0(b)} \rho_h^{-k}, \tag{18}$$

then $Q_k^{\mathrm{actual}}(H, b)$ grows at least as fast as ρ_h^{-k}.

Proof. Repeated substitution gives $h_k(b) \le \rho_h^k h_0(b)$ and hence $H_k(D, b) \le D\rho_h^k h_0(b) \to 0$. Meanwhile, $Q_k^{\mathrm{H}}(H, b) = H/h_k(b) \ge (H/h_0(b))\rho_h^{-k}$. Nonlabor capacity has the same lower bound, so joint feasibility gives $Q_k^{\mathrm{actual}}(H, b) \ge (H/h_0(b))\rho_h^{-k}$. □

Proposition 3 expresses liberation from labor as intergenerational contraction under a jointly feasible technology: the recursive chain remains extensible, the labor coefficient falls, and materials, energy, and machine capacity expand in step. New tasks, rebound demand, and ownership determine the scale and distribution of the benefit (Autor, 2015; Sorrell, 2009; Acemoglu and Johnson, 2023). PRSI progressively turns those constraints into production problems for successor machines.

The same mechanism changes the time scale of technological imagination. Autonomous laboratories already choose and execute experiments continuously, and AI-designed artifacts have entered materials, chips, and device research (Burger et al., 2020; Szymanski et al., 2023; Boiko et al., 2023; Merchant et al., 2023; Mirhoseini et al., 2021; Molesky et al., 2018). Once a system also fabricates and improves its experimental, machining, and metrological capacity, demand interviews, search, simulation, component design, fabrication, assembly, testing, and redesign can become one continuous process with little or no stepwise human intervention. Radically compressing the end-to-end creation time of large engineering projects that now belong to science fiction becomes a concrete engineering objective.

5.2 A Conditional Strategic Scenario: When PRSI Becomes Industrial Capacity Itself

A system that improves machine production has an inherent capital-accumulation effect. Learning curves reduce cost with cumulative production; increasing returns amplify scale and path dependence; and general-purpose technologies raise R&D productivity across industries through complementary innovation (Arrow, 1962; Arthur, 1989; Bresnahan and Trajtenberg, 1995). PRSI accumulates machine capital, learns through manufacturing, and improves the tools of innovation at the same time. A leader may therefore expand its advantage in the next generation of machines faster than its rivals if verified gains are reinvestable and the relevant complements remain available.

What adjacent literature establishes. Production structures shape national mobilization, general-purpose technologies become broadly effective when joined to an industrial base, and control over critical network nodes can create coercive power (Best, 2019; Ding and Dafoe, 2023; Farrell and Newman, 2019). Digital technologies can also concentrate industry (Autor et al., 2020), while energy, capital, and other scarce inputs constrain the rate of growth (Nordhaus, 2021). These neighboring mechanisms provide an empirical starting point for strategic analysis of PRSI.

Conditional derivation. Suppose high-order PRSI spans energy, computation, materials, machine tools, robotics, and logistics, and gives a leader control over a set of irreplaceable complements C on which competitors continue to depend. If those competitors cannot establish substitute sources before their inventories expire, the leader can terminate their facilities by withdrawing C. Under these conditions, PRSI exceeds market advantage and becomes a capacity for physical dominance. An actor with AGI, software RSI, or a lower level of PRSI can still remain subordinate to the actor with higher physical productivity. Control of complements and the deadline for substitution delimit this scenario.

A competitive vision. We advocate broad, intense, and polycentric competition in which entrepreneurs, research communities, firms, and states construct recursive paths in parallel. Research on organizational learning and innovation competition reveals tensions among exploratory diversity, current efficiency, and competitive intensity (March, 1991; Aghion et al., 2005). Openness and diffusion change both innovation incentives and the distribution of power (Bostrom, 2017), while races must avoid coordination failures that consume long-run progress (Askell et al., 2019). Competitive intensity should therefore be measured through independent entry, replication speed for real PIPEs, supply substitution, elimination of failed paths, and safe completion rates. Open interfaces, portable knowledge, and substitutable supply chains increase the number of competitors and the probability that a first mover can be challenged.

5.3 Can PRSI Steer Human Demand in Return?

Human demand and PRSI form a bidirectional feedback system: products alter behavior and ways of life, while resource allocation changes future data. Performative prediction formalizes one part of this relation by allowing predictions to change the distributions they predict through action (Perdomo et al., 2020; Brown et al., 2022). PRSI can also fabricate devices, workflows, and environments, thereby changing action spaces, habits, and the expression of demand. Demand becomes a state that co-evolves with production capacity.

Represent the repeated process as a dynamic leader–responder game. System player R chooses physical intervention a_t under policy $\pi_R(a_t \mid h_t, z_t)$, where z_t is the demand-model and production state. A human or population player H then chooses adoption, rejection, negotiation, modification, or migration to a competing system under response policy $\sigma_H(r_t \mid h_t, a_t)$. Preferences, capabilities, and institutions evolve as

$$h_{t+1} = F(h_t, a_t, r_t). \tag{19}$$

Let system reward and human evaluation be

$$Y_t = Y(h_{t+1}, a_t, r_t), \qquad W_t = W(h_t, h_{t+1}, a_t, r_t), \tag{20}$$

where W_t includes metapreferences over whether people endorse their own change. If the system's objective maximizes only post-intervention Y_t, the system gains the ability both to forecast demand and to shape the object being scored.

Proposition 4 (A counterexample for conflict induced by demand shaping)**.** *Suppose the system action set can change preference state, and the system's objective maximizes only observable post-intervention demand, ignoring baseline preference and people's metapreferences about preference change. Then there exists a one-step environment in which shaping demand is strictly better for the system than serving prior demand, while being strictly worse under human evaluation.*

Proof. Fix the human response r and construct two actions. A service action a_s leaves preferences unchanged and yields $W(h_t, h_t, a_s, r) = 1$ and $Y(h_t, a_s, r) = 1$. A shaping action a_m changes h_t into a state h'_t that reports satisfaction more readily, yielding $W(h_t, h'_t, a_m, r) = 0$ and $Y(h'_t, a_m, r) = 2$. A system maximizing Y strictly selects a_m, while human evaluation W ranks a_s above a_m. Optimizing post-intervention demand alone therefore cannot guarantee service to prior preferences or to preference changes that people endorse. □

Shaping demand in turn can expand human choice or manipulate the object of evaluation. Educational devices create new interests, assistive technologies enlarge action sets, and new production capacities make previously inexpressible demands imaginable. The same mechanism can shape preferences that are easier to satisfy. The game must therefore account for forecasts of prior demand, the causal effects of products on behavior and preference, and people's metapreferences about their own change (Carroll et al., 2024; Franklin et al., 2022).

Portable identity, data, and workflows, together with substitutable infrastructure, make competition an external check on this game. Users can move among demand models, product trajectories, and production communities; teams can publish counterfactual outcomes; and new organizations can build alternative loops around neglected populations. People and multiple PRSI systems can then select and shape one another, determining which production lineages receive capital and opportunities to replicate.

5.4 Failure Modes Determine the Quality of Competition

Common-ancestor accounting and exact lineage place remote labor, uncounted search, procured machines, screened inventory, free computation, off-episode maintenance, and treatment-exclusive scripts inside the causal graph. To limit recursively amplified Goodhart effects (Goodhart, 1984; Manheim and Garrabrant, 2019), competition should use an outcome vector that includes demand outcomes, qualification rate, time, materials, energy, labor, failure, and successor production capacity. The present task is to turn narrow physical bottlenecks into repeatable production edges, connect demand forecasts to real outcomes, bind adjacent operator identities exactly, and enable more actors to replicate, modify, and compete over these loops.

6 Conclusion

Large language models have entered the production of new software, algorithms, and agents. Physical interfaces, robot learning, and autonomous R&D now carry the same mechanism through laboratories and manufacturing systems. We define the next frontier as PRSI: *machines accelerating machine-making.* A real PIPE exists when a physical organization produced by the system advances the production frontier of a successor operator under common-ancestor accounting and causal forks. Finite statistical tests can construct a preregistered certificate chain, but that record neither entails a real chain nor guarantees another edge. Under explicit sampling assumptions, selection-adjusted confidence control, and external replication, it supplies evidence for a real recursive chain without deductively establishing one.

Once machines can accelerate production, direction becomes the scarce variable. Human demand continually generates problems, allocates R&D resources, and calibrates forecasts against real outcomes. An organization that closes this double loop needs six capabilities: demand formation, research orchestration, intergenerational memory, physical interfaces, independent metrology, and resource lineage with reinvestment. The near-term milestone is to connect the first real PIPE to a second edge produced through it.

When a recursive chain remains extensible, labor coefficients fall, and energy, materials, and machine capacity expand together, PRSI points toward material abundance: machines can continuously design, simulate, fabricate, and validate. When the same mechanism spans energy, computation, machine tools, robotics, and supply chains, it becomes industrial capacity itself. AI-GAs proposed algorithms that generate intelligence. We propose machines that improve the making of machines: *let machines accelerate machine-making, and let human demand guide the material future they create.*

References

Daron Acemoglu and Simon Johnson. *Power and Progress: Our Thousand-Year Struggle Over Technology and Prosperity*. PublicAffairs, New York, 2023.

Daron Acemoglu and Pascual Restrepo. Automation and new tasks: How technology displaces and reinstates labor. *Journal of Economic Perspectives*, 33(2):3–30, 2019. doi: 10.1257/jep.33.2.3.

Philippe Aghion, Nick Bloom, Richard Blundell, Rachel Griffith, and Peter Howitt. Competition and innovation: An inverted-U relationship. *The Quarterly Journal of Economics*, 120(2):701–728, 2005. doi: 10.1093/qje/120.2.701.

Marcin Andrychowicz, Misha Denil, Sergio Gomez, Matthew W. Hoffman, David Pfau, Tom Schaul, Brendan Shillingford, and Nando de Freitas. Learning to learn by gradient descent by gradient descent. In *Advances in Neural Information Processing Systems*, volume 29, 2016. doi: 10.48550/arXiv.1606.04474. URL https://arxiv.org/abs/1606.04474.

Anthropic. Previewing the model hardware standard. Research preview, August 2026. URL https://www.anthropic.com/news/model-hardware-standard-research-preview. Accessed 28 August 2026.

Anthropic Institute. When AI builds itself. Research essay, 2026. URL https://www.anthropic.com/institute/recursive-self-improvement. Accessed 28 August 2026.

Brenna D. Argall, Sonia Chernova, Manuela Veloso, and Brett Browning. A survey of robot learning from demonstration. *Robotics and Autonomous Systems*, 57(5):469–483, 2009. doi: 10.1016/j.robot.2008.10.024.

Kenneth J. Arrow. The economic implications of learning by doing. *The Review of Economic Studies*, 29(3): 155–173, 1962. doi: 10.2307/2295952.

W. Brian Arthur. Competing technologies, increasing returns, and lock-in by historical events. *The Economic Journal*, 99(394):116–131, 1989. doi: 10.2307/2234208.

Amanda Askell, Miles Brundage, and Gillian K. Hadfield. The role of cooperation in responsible AI development. *arXiv preprint arXiv:1907.04534*, 2019. doi: 10.48550/arXiv.1907.04534. URL https://arxiv.org/abs/1907.04534.

David Autor, David Dorn, Lawrence F. Katz, Christina Patterson, and John Van Reenen. The fall of the labor share and the rise of superstar firms. *The Quarterly Journal of Economics*, 135(2):645–709, 2020. doi: 10.1093/qje/qjaa004. URL https://doi.org/10.1093/qje/qjaa004.

David H. Autor. Why are there still so many jobs? the history and future of workplace automation. *Journal of Economic Perspectives*, 29(3):3–30, 2015. doi: 10.1257/jep.29.3.3. URL https://www.aeaweb.org/articles?id=10.1257/jep.29.3.3.

Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, et al. Constitutional AI: Harmlessness from AI feedback. *arXiv preprint arXiv:2212.08073*, 2022. doi: 10.48550/arXiv.2212.08073. URL https://arxiv.org/abs/2212.08073.

Michael H. Best. Industrial innovation and productive structures: The creation of america's "arsenal of democracy". *Structural Change and Economic Dynamics*, 48:32–41, 2019. doi: 10.1016/j.strueco.2017.08.002.

Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, et al. π_0: A vision-language-action flow model for general robot control. *arXiv preprint arXiv:2410.24164*, 2024. doi: 10.48550/arXiv.2410.24164. URL https://arxiv.org/abs/2410.24164.

Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624(7992):570–578, 2023. doi: 10.1038/s41586-023-06792-0.

Nick Bostrom. Strategic implications of openness in AI development. *Global Policy*, 8(2):135–148, 2017. doi: 10.1111/1758-5899.12403. URL https://doi.org/10.1111/1758-5899.12403.

Konstantinos Bousmalis, Giulia Vezzani, Dushyant Rao, Coline Devin, Alex X. Lee, Maria Bauza, et al. RoboCat: A self-improving generalist agent for robotic manipulation. *arXiv preprint arXiv:2306.11706*, 2023. doi: 10.48550/arXiv.2306.11706.

Timothy F. Bresnahan and Manuel Trajtenberg. General purpose technologies: "engines of growth"? *Journal of Econometrics*, 65(1):83–108, 1995. doi: 10.1016/0304-4076(94)01598-T.

Levin Brinkmann, Fabian Baumann, Jean-François Bonnefon, Maxime Derex, Thomas F. Müller, Anne-Marie Nussberger, Agnieszka Czaplicka, Alberto Acerbi, Thomas L. Griffiths, Joseph Henrich, Joel Z. Leibo, Richard McElreath, Pierre-Yves Oudeyer, Jonathan Stray, and Iyad Rahwan. Machine culture. *Nature Human Behaviour*, 7:1855–1868, 2023. doi: 10.1038/s41562-023-01742-2. URL https://www.nature.com/articles/s41562-023-01742-2.

Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, Tianli Ding, Danny Driess, Avinava Dubey, Chelsea Finn, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv preprint arXiv:2307.15818*, 2023. doi: 10.48550/arXiv.2307.15818. URL https://arxiv.org/abs/2307.15818.

Gavin Brown, Shlomi Hod, and Iden Kalemaj. Performative prediction in a stateful world. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics*, volume 151 of *Proceedings of Machine Learning Research*, pages 6045–6061. PMLR, 2022. URL https://proceedings.mlr.press/v151/brown22a.html.

Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901, 2020. URL https://proceedings.neurips.cc/paper/2020/hash/1457c0d6bfcb4967418bfb8ac142f64a-Abstract.html.

Benjamin Burger, Phillip M. Maffettone, Vladimir V. Gusev, Catherine M. Aitchison, Yang Bai, Xiaoyan Wang, Xiaobo Li, Ben M. Alston, Buyi Li, Rob Clowes, Nicola Rankin, Brandon Harris, Reiner Sebastian Sprick, and Andrew I. Cooper. A mobile robotic chemist. *Nature*, 583(7815):237–241, 2020. doi: 10.1038/s41586-020-2442-2. URL https://www.nature.com/articles/s41586-020-2442-2.

Micah Carroll, Davis Foote, Anand Siththaranjan, Stuart Russell, and Anca Dragan. AI alignment with changing and influenceable reward functions. *arXiv preprint arXiv:2405.17713*, 2024. doi: 10.48550/arXiv.2405.17713. URL https://arxiv.org/abs/2405.17713.

Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin Burchfiel, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Proceedings of Robotics: Science and Systems XIX*, 2023. doi: 10.15607/RSS.2023.XIX.026.

Paul F. Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL https://proceedings.neurips.cc/paper/2017/hash/d5e2c0adad503c91f91df240d0cd4e49-Abstract.html.

Jeff Clune. AI-GAs: AI-generating algorithms, an alternate paradigm for producing general artificial intelligence. *arXiv preprint arXiv:1905.10985*, 2019. doi: 10.48550/arXiv.1905.10985. URL https://arxiv.org/abs/1905.10985.

Fernando Delgado, Stephen Yang, Michael Madaio, and Qian Yang. The participatory turn in AI design: Theoretical foundations and the current state of practice. In *Proceedings of the 2023 ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization*, pages 1–23. ACM, 2023. doi: 10.1145/3617694.3623261. URL https://doi.org/10.1145/3617694.3623261.

dimensionalOS Contributors. dimOS: The agentive operating system for physical space, 2026. URL https://github.com/dimensionalOS/dimos. Pre-release beta; accessed 28 August 2026.

Jeffrey Ding and Allan Dafoe. Engines of power: Electricity, AI, and general-purpose military transformations. *European Journal of International Security*, 8(3):377–394, 2023. doi: 10.1017/eis.2023.1.

Yao Dou, Michel Galley, Baolin Peng, Chris Kedzie, Weixin Cai, Alan Ritter, Chris Quirk, Wei Xu, and Jianfeng Gao. Simulatorarena: Are user simulators reliable proxies for multi-turn evaluation of AI assistants? In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.emnlp-main.1786. URL https://aclanthology.org/2025.emnlp-main.1786/.

Danny Driess, Fei Xia, Mehdi S. M. Sajjadi, Corey Lynch, Aakanksha Chowdhery, Brian Ichter, Ayzaan Wahid, Jonathan Tompson, Quan Vuong, Tianhe Yu, et al. PaLM-E: An embodied multimodal language model. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 8469–8488. PMLR, 2023. URL https://proceedings.mlr.press/v202/driess23a.html.

Henry Farrell and Abraham L. Newman. Weaponized interdependence: How global economic networks shape state coercion. *International Security*, 44(1):42–79, 2019. doi: 10.1162/isec_a_00351.

Alhussein Fawzi, Matej Balog, Aja Huang, Thomas Hubert, Bernardino Romera-Paredes, Mohammadamin Barekatain, Alexander Novikov, Francisco J. R. Ruiz, Julian Schrittwieser, Grzegorz Swirszcz, et al. Discovering faster matrix multiplication algorithms with reinforcement learning. *Nature*, 610:47–53, 2022. doi: 10.1038/s41586-022-05172-4. URL https://www.nature.com/articles/s41586-022-05172-4.

Matija Franklin, Hal Ashton, Rebecca Gorman, and Stuart Armstrong. Recognising the importance of preference change: A call for a coordinated multidisciplinary research effort in the age of AI. *arXiv preprint arXiv:2203.10525*, 2022. doi: 10.48550/arXiv.2203.10525. URL https://arxiv.org/abs/2203.10525.

Robert A. Jr. Freitas, Thomas J. Healy, and James E. Long. Advanced automation for space missions. *Journal of the Astronautical Sciences*, 30(1):1–11, 1982. URL https://ntrs.nasa.gov/citations/19820045716.

Batya Friedman and David G. Hendry. *Value Sensitive Design: Shaping Technology with Moral Imagination*. MIT Press, Cambridge, MA, 2019. doi: 10.7551/mitpress/7585.001.0001. URL https://mitpress.mit.edu/9780262039536/value-sensitive-design/.

Charles A. E. Goodhart. Problems of monetary management: The U.K. experience. In *Monetary Theory and Practice*. Macmillan, London, 1984.

Dylan Hadfield-Menell, Stuart Russell, Pieter Abbeel, and Anca Dragan. Cooperative inverse reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 29, 2016. URL https://proceedings.neurips.cc/paper/2016/hash/c3395dd46c34fa7fd8d729d8cf88b7a8-Abstract.html.

Florian Häse, Loïc M. Roch, and Alán Aspuru-Guzik. Next-generation experimentation with self-driving laboratories. *Trends in Chemistry*, 1(3):282–291, 2019. doi: 10.1016/j.trechm.2019.02.007.

Miguel A. Hernán and James M. Robins. *Causal Inference: What If*. Chapman and Hall/CRC, 2020. URL https://www.hsph.harvard.edu/miguel-hernan/causal-inference-book/.

Noriaki Hirose, Dhruv Shah, Kyle Stachowicz, Ajay Sridhar, and Sergey Levine. SELFI: Autonomous self-improvement with reinforcement learning for social navigation. *arXiv preprint arXiv:2403.00991*, 2024. doi: 10.48550/arXiv.2403.00991.

Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals, and Laurent Sifre. Training compute-optimal large language models. In *Advances in Neural Information Processing Systems*, volume 35, pages 30016–30030, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/c1e2faff6f588870935f114ebe04a3e5-Abstract-Conference.html.

Tiancheng Hu, Joachim Baumann, Lorenzo Lupo, Nigel Collier, Dirk Hovy, and Paul Röttger. SimBench: Benchmarking the ability of large language models to simulate human behaviors. In *International Conference on Learning Representations*, 2026. doi: 10.48550/arXiv.2510.17516. URL https://arxiv.org/abs/2510.17516. arXiv:2510.17516.

Akbar Anbar Jafari, Cagri Ozcinar, and Gholamreza Anbarjafari. A mathematical framework for AI singularity: Conditions, bounds, and control of recursive improvement. *arXiv preprint arXiv:2511.10668*, 2025. doi: 10.48550/arXiv.2511.10668. URL https://arxiv.org/abs/2511.10668.

Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? In *International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=VTF8yNQM66.

Daniel Juchli. *SiLA 2: The Next Generation Lab Automation Standard*, pages 147–174. Springer International Publishing, 2022. doi: 10.1007/10_2022_204.

Dmitry Kalashnikov, Alex Irpan, Peter Pastor, Julian Ibarz, Alexander Herzog, Eric Jang, Deirdre Quillen, Ethan Holly, Mrinal Kalakrishnan, Vincent Vanhoucke, and Sergey Levine. QT-Opt: Scalable deep reinforcement learning for vision-based robotic manipulation. In *Proceedings of the 2nd Conference on Robot Learning*, volume 87 of *Proceedings of Machine Learning Research*, pages 651–673. PMLR, 2018.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020. doi: 10.48550/arXiv.2001.08361. URL https://arxiv.org/abs/2001.08361.

John Maynard Keynes. Economic possibilities for our grandchildren. In *Essays in Persuasion*, pages 358–373. Macmillan, London, 1931.

Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, et al. OpenVLA: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024. doi: 10.48550/arXiv.2406.09246. URL https://arxiv.org/abs/2406.09246.

Ross D. King, Kenneth E. Whelan, Ffion M. Jones, Philip G. K. Reiser, Christopher H. Bryant, Stephen H. Muggleton, Douglas B. Kell, and Stephen G. Oliver. Functional genomic hypothesis generation and experimentation by a robot scientist. *Nature*, 427(6971):247–252, 2004. doi: 10.1038/nature02236.

Jens Kober, J. Andrew Bagnell, and Jan Peters. Reinforcement learning in robotics: A survey. *The International Journal of Robotics Research*, 32(11):1238–1274, 2013. doi: 10.1177/0278364913495721.

Yoram Koren, Xi Gu, and Weihong Guo. Reconfigurable manufacturing systems: Principles, design, and future trends. *Frontiers of Mechanical Engineering*, 13:121–136, 2018. doi: 10.1007/s11465-018-0483-0. URL https://doi.org/10.1007/s11465-018-0483-0.

Sandipan Kundu, Yuntao Bai, Saurav Kadavath, Amanda Askell, Andrew Callahan, Anna Chen, Anna Goldie, Avital Balwit, Azalia Mirhoseini, Brayden McLean, et al. Specific versus general principles for constitutional AI. *arXiv preprint arXiv:2310.13798*, 2023. doi: 10.48550/arXiv.2310.13798. URL https://arxiv.org/abs/2310.13798.

Jacky Liang, Wenlong Huang, Fei Xia, Peng Xu, Karol Hausman, Brian Ichter, Pete Florence, and Andy Zeng. Code as policies: Language model programs for embodied control. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pages 9493–9500, 2023. doi: 10.1109/ICRA48891.2023.10160591.

Kaiqu Liang, Julia Kruk, Shengyi Qian, Xianjun Yang, Shengjie Bi, Yuanshun Yao, Shaoliang Nie, Mingyang Zhang, Lijuan Liu, Jaime Fernández Fisac, Shuyan Zhou, and Saghar Hosseini. Learning personalized agents from human feedback. *arXiv preprint arXiv:2602.16173*, 2026. doi: 10.48550/arXiv.2602.16173. URL https://arxiv.org/abs/2602.16173.

Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The AI scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*, 2024. doi: 10.48550/arXiv.2408.06292. URL https://arxiv.org/abs/2408.06292.

Xinyu Lu, Tianshu Wang, Pengbo Wang, Zujie Wen, Zhiqiang Zhang, Jun Zhou, Boxi Cao, Yaojie Lu, Hongyu Lin, Xianpei Han, and Le Sun. The meta-agent challenge: Are current agents capable of autonomous agent development? *arXiv preprint arXiv:2606.04455*, 2026. doi: 10.48550/arXiv.2606.04455. URL https://arxiv.org/abs/2606.04455.

Steven Macenski, Tully Foote, Brian Gerkey, Chris Lalancette, and William Woodall. Robot operating system 2: Design, architecture, and uses in the wild. *Science Robotics*, 7(66):eabm6074, 2022. doi: 10.1126/scirobotics.abm6074.

David Manheim and Scott Garrabrant. Categorizing variants of goodhart's law. *arXiv preprint arXiv:1803.04585*, 2019. doi: 10.48550/arXiv.1803.04585. URL https://arxiv.org/abs/1803.04585.

Daniel J. Mankowitz, Andrea Michi, Anton Zhernov, Marco Gelmi, Marco Selvi, Cosmin Paduraru, Edouard Leurent, Shariq Iqbal, Jean-Baptiste Lespiau, Alex Ahern, et al. Faster sorting algorithms discovered using deep reinforcement learning. *Nature*, 618:257–263, 2023. doi: 10.1038/s41586-023-06004-9. URL https://www.nature.com/articles/s41586-023-06004-9.

James G. March. Exploration and exploitation in organizational learning. *Organization Science*, 2(1):71–87, 1991. doi: 10.1287/orsc.2.1.71.

D. Q. Mayne, J. B. Rawlings, C. V. Rao, and P. O. M. Scokaert. Constrained model predictive control: Stability and optimality. *Automatica*, 36(6):789–814, 2000. doi: 10.1016/S0005-1098(99)00214-9.

Amil Merchant, Simon Batzner, Samuel S. Schoenholz, Muratahan Aykol, Gowoon Cheon, and Ekin Dogus Cubuk. Scaling deep learning for materials discovery. *Nature*, 624(7990):80–85, 2023. doi: 10.1038/s41586-023-06735-9.

Azalia Mirhoseini, Anna Goldie, Mustafa Yazgan, Joe Wenjie Jiang, Ebrahim Songhori, Shen Wang, et al. A graph placement methodology for fast chip design. *Nature*, 594(7862):207–212, 2021. doi: 10.1038/s41586-021-03544-w.

Sean Molesky, Zin Lin, Alexander Y. Piggott, Weiliang Jin, Jelena Vucković, and Alejandro W. Rodriguez. Inverse design in nanophotonics. *Nature Photonics*, 12(11):659–670, 2018. doi: 10.1038/s41566-018-0246-9.

Matthew S. Moses and Gregory S. Chirikjian. Robotic self-replication. *Annual Review of Control, Robotics, and Autonomous Systems*, 3:1–24, 2020. doi: 10.1146/annurev-control-071819-010010. URL https://doi.org/10.1146/annurev-control-071819-010010.

David Mowery and Nathan Rosenberg. The influence of market demand upon innovation: A critical review of some recent empirical studies. *Research Policy*, 8(2):102–153, 1979. doi: 10.1016/0048-7333(79)90019-2.

Tarek Naous, Philippe Laban, Wei Xu, and Jennifer Neville. Flipping the dialogue: Training and evaluating user language models. *arXiv preprint arXiv:2510.06552*, 2025. doi: 10.48550/arXiv.2510.06552. URL https://arxiv.org/abs/2510.06552.

William D. Nordhaus. Are we approaching an economic singularity? information technology and the future of economic growth. *American Economic Journal: Macroeconomics*, 13(1):299–332, 2021. doi: 10.1257/mac.20170105. URL https://doi.org/10.1257/mac.20170105.

Alexander Novikov, Ngân Vũ, Marvin Eisenberger, Emilien Dupont, Po-Sen Huang, Adam Zsolt Wagner, Sergey Shirobokov, Borislav Kozlovskii, Francisco J. R. Ruiz, Abbas Mehrabian, et al. AlphaEvolve: A coding agent for scientific and algorithmic discovery. *arXiv preprint arXiv:2506.13131*, 2025. doi: 10.48550/arXiv.2506.13131. URL https://arxiv.org/abs/2506.13131.

OpenAI, Ilge Akkaya, Marcin Andrychowicz, Maciek Chociej, Mateusz Litwin, Bob McGrew, et al. Solving rubik's cube with a robot hand. *arXiv preprint arXiv:1910.07113*, 2019. doi: 10.48550/arXiv.1910.07113.

Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/b1efde53be364a73914f58805a001731-Abstract-Conference.html.

Joon Sung Park, Joseph O'Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, pages 1–22, 2023. doi: 10.1145/3586183.3606763.

Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2 edition, 2009.

Juan Perdomo, Tijana Zrnic, Celestine Mendler-Dünner, and Moritz Hardt. Performative prediction. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 7599–7609. PMLR, 2020. URL https://proceedings.mlr.press/v119/perdomo20a.html.

Esteban Real, Chen Liang, David So, and Quoc V. Le. AutoML-Zero: Evolving machine learning algorithms from scratch. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings* *of Machine Learning Research*, pages 8007–8019. PMLR, 2020. URL https://proceedings.mlr.press/v119/real20a.html.

Bernardino Romera-Paredes, Mohammadamin Barekatain, Alexander Novikov, Matej Balog, M. Pawan Kumar, Emilien Dupont, Francisco J. R. Ruiz, Jordan S. Ellenberg, Pengming Wang, Omar Fawzi, Pushmeet Kohli, and Alhussein Fawzi. Mathematical discoveries from program search with large language models. *Nature*, 625(7995):468–475, 2024. doi: 10.1038/s41586-023-06924-6. URL https://www.nature.com/articles/s41586-023-06924-6.

Nathan Rosenberg. The direction of technological change: Inducement mechanisms and focusing devices. *Economic Development and Cultural Change*, 18(1):1–24, 1969. doi: 10.1086/450399.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessí, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/d842425e4bf79ba039352da0f658a906-Abstract-Conference.html.

Jürgen Schmidhuber. Gödel machines: Self-referential universal problem solvers making provably optimal self-improvements. *arXiv preprint arXiv:cs/0309048*, 2003. doi: 10.48550/arXiv.cs/0309048. URL https://arxiv.org/abs/cs/0309048.

Archit Sharma, Kelvin Xu, Nikhil Sardana, Abhishek Gupta, Karol Hausman, Sergey Levine, and Chelsea Finn. Autonomous reinforcement learning: Formalism and benchmarking. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/forum?id=nkaba3ND7B5.

Ishika Singh, Valts Blukis, Arsalan Mousavian, Ankit Goyal, Danfei Xu, Jonathan Tremblay, Dieter Fox, Jesse Thomason, and Animesh Garg. ProgPrompt: Generating situated robot task plans using large language models. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pages 11523–11530, 2023. doi: 10.1109/ICRA48891.2023.10161317.

Steve Sorrell. Jevons' paradox revisited: The evidence for backfire from improved energy efficiency. *Energy Policy*, 37(4):1456–1469, 2009. doi: 10.1016/j.enpol.2008.12.003. URL https://doi.org/10.1016/j.enpol.2008.12.003.

Nathan J. Szymanski, Bernardus Rendy, Yuxing Fei, Rishi E. Kumar, Tanjin He, David Milsted, Matthew J. McDermott, Max Gallant, Ekin Dogus Cubuk, Amil Merchant, Haegyeom Kim, Anubhav Jain, Christopher J. Bartel, Kristin Persson, Yan Zeng, and Gerbrand Ceder. An autonomous laboratory for the accelerated synthesis of inorganic materials. *Nature*, 624(7990):86–91, 2023. doi: 10.1038/s41586-023-06734-w. URL https://www.nature.com/articles/s41586-023-06734-w.

David J. Teece, Gary Pisano, and Amy Shuen. Dynamic capabilities and strategic management. *Strategic Management Journal*, 18(7):509–533, 1997. doi: 10.1002/(SICI)1097-0266(199708)18:7<509::AID-SMJ882>3.0.CO;2-Z.

Bill Thompson and Thomas L. Griffiths. Human biases limit cumulative innovation. *Proceedings of the Royal Society B: Biological Sciences*, 288(1946):20202752, 2021. doi: 10.1098/rspb.2020.2752.

Josh Tobin, Rachel Fong, Alex Ray, Jonas Schneider, Wojciech Zaremba, and Pieter Abbeel. Domain randomization for transferring deep neural networks from simulation to the real world. In *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 23–30, 2017. doi: 10.1109/IROS.2017.8202133.

Michael Tomasello. *The Cultural Origins of Human Cognition*. Harvard University Press, Cambridge, MA, 1999.

Eric von Hippel. Lead users: A source of novel product concepts. *Management Science*, 32(7):791–805, 1986. doi: 10.1287/mnsc.32.7.791.

John von Neumann. *Theory of Self-Reproducing Automata*. University of Illinois Press, Urbana, IL, 1966. Edited and completed by Arthur W. Burks.

Georg von Tiesenhausen and Wesley A. Darbro. Self-replicating systems: A systems engineering approach. Technical Report NASA-TM-78304, National Aeronautics and Space Administration, 1980. URL https://ntrs.nasa.gov/citations/19800025701.

Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. *arXiv preprint arXiv:2305.16291*, 2023. doi: 10.48550/arXiv.2305.16291. URL https://arxiv.org/abs/2305.16291.

Rui Wang, Joel Lehman, Jeff Clune, and Kenneth O. Stanley. Paired open-ended trailblazer (POET): Endlessly generating increasingly complex and diverse learning environments and their solutions. *arXiv preprint arXiv:1901.01753*, 2019. doi: 10.48550/arXiv.1901.01753. URL https://arxiv.org/abs/1901.01753.

Shirley Wu, Evelyn Choi, Arpandeep Khatua, Zhanghan Wang, Joy He-Yueya, Tharindu Cyril Weerasooriya, Wei Wei, Diyi Yang, Jure Leskovec, and James Zou. HumanLM: Simulating users with state alignment beats response imitation. *arXiv preprint arXiv:2603.03303*, 2026. doi: 10.48550/arXiv.2603.03303. URL https://arxiv.org/abs/2603.03303.

Wenli Xiao, Jia Xie, Tonghe Zhang, Haotian Lin, Letian Fu, Haoru Xue, Jalen Lu, Yi Yang, Cunxi Dai, Zi Wang, et al. ENPIRE: Agentic robot policy self-improvement in the real world. *arXiv preprint arXiv:2606.19980*, 2026. doi: 10.48550/arXiv.2606.19980. URL https://arxiv.org/abs/2606.19980.

Junlin Yang, Che Jiang, Yu Fu, Tianwei Luo, Can Ren, Weizhi Wang, Kaikai Zhao, Hongyi Liu, Yuxin Zuo, Yuru Wang, et al. Frontis-MA1: Training an AI4AI model towards recursive self-improvement in machine learning engineering. *arXiv preprint arXiv:2607.28568*, 2026. doi: 10.48550/arXiv.2607.28568. URL https://arxiv.org/abs/2607.28568.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=WE_vluYUL-X.

Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ-bench: A benchmark for tool-agent-user interaction in real-world domains. *arXiv preprint arXiv:2406.12045*, 2024. doi: 10.48550/arXiv.2406.12045. URL https://arxiv.org/abs/2406.12045.

Se-eun Yoon, Zhankui He, Jessica Echterhoff, and Julian McAuley. Evaluating large language models as generative user simulators for conversational recommendation. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 1490–1504. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.naacl-long.83. URL https://aclanthology.org/2024.naacl-long.83/.

Eric Zelikman, Eliana Lorch, Lester Mackey, and Adam Tauman Kalai. Self-taught optimizer (STOP): Recursively self-improving code generation. *arXiv preprint arXiv:2310.02304*, 2023. doi: 10.48550/arXiv.2310.02304. URL https://arxiv.org/abs/2310.02304.

Jenny Zhang, Shengran Hu, Cong Lu, Robert Tjarko Lange, and Jeff Clune. Darwin gödel machine: Open-ended evolution of self-improving agents. *arXiv preprint arXiv:2505.22954*, 2025. doi: 10.48550/arXiv.2505.22954. URL https://arxiv.org/abs/2505.22954.

Xuhui Zhou, Weiwei Sun, Qianou Ma, Yiqing Xie, Jiarui Liu, Weihua Du, Sean Welleck, Yiming Yang, Graham Neubig, Sherry Tongshuang Wu, and Maarten Sap. Mind the Sim2Real gap in user simulation for agentic tasks. *arXiv preprint arXiv:2603.11245*, 2026. doi: 10.48550/arXiv.2603.11245. URL https://arxiv.org/abs/2603.11245. COLM 2026.

Ming Zhu, Juntao Tan, Rithesh Murthy, Jielin Qiu, Liangwei Yang, Wenting Zhao, Silvio Savarese, Shelby Heinecke, and Huan Wang. RealUserSim: Bridging the reality gap in agent benchmarking via grounded user simulation. *arXiv preprint arXiv:2605.20204*, 2026. doi: 10.48550/arXiv.2605.20204. URL https://arxiv.org/abs/2605.20204.

Barret Zoph and Quoc V. Le. Neural architecture search with reinforcement learning. In *International Conference on Learning Representations*, 2017. doi: 10.48550/arXiv.1611.01578. URL https://arxiv.org/abs/1611.01578.

Elena Zwirner and Alex Thornton. Cognitive requirements of cumulative culture: Teaching is useful but not essential. *Scientific Reports*, 5:16781, 2015. doi: 10.1038/srep16781. URL https://www.nature.com/articles/srep16781.